

Repetitive Impedance Learning-Based Physically Human–Robot Interactive Control

Tairen Sun¹, Jiantao Yang¹, Yongping Pan¹, *Senior Member, IEEE*, and Hongliu Yu¹

Abstract—Model-based impedance learning control can provide variable impedance regulation for robots through online impedance learning without interaction force sensing. However, the existing related results only guarantee the closed-loop control systems to be uniformly ultimately bounded (UUB) and require the human impedance profiles being periodic, iteration-dependent, or slowly varying. In this article, a repetitive impedance learning control approach is proposed for physical human–robot interaction (PHRI) in repetitive tasks. The proposed control is composed of a proportional-differential (PD) control term, an adaptive control term, and a repetitive impedance learning term. Differential adaptation with projection modification is designed for estimating robotic parameters uncertainties in the time domain, while fully saturated repetitive learning is proposed for estimating time-varying human impedance uncertainties in the iterative domain. Uniform convergence of tracking errors is guaranteed by the PD control and the use of projection and full saturation in the uncertainties estimation and is theoretically proved based on a Lyapunov-like analysis. In impedance profiles, the stiffness and damping are composed of an iteration-independent term and an iteration-dependent disturbance, which are estimated by repetitive learning and compressed by the PD control, respectively. Therefore, the developed approach can be applied to the PHRI where iteration-dependent disturbances exist in the stiffness and damping. The control effectiveness and advantages are validated by simulations on a parallel robot in a repetitive following task.

Index Terms—Impedance control, impedance learning, iterative learning, physical human–robot interaction (PHRI), repetitive learning control, robot control.

I. INTRODUCTION

SAFETY is the first concern for physical human–robot interaction (PHRI) and can be improved by robotic compliance control through robot impedance regulation. As the most widely used compliance control approach, impedance control can provide robot impedance regulation by the

desired spring-damping dynamics relationship between interaction forces and tracking errors [1], [2], [3]. To implement impedance control for PHRI, there are two significant problems to be solved. One is impedance control stability, and the other is human impedance learning.

For impedance control, the stability of the desired impedance dynamics and impedance errors should be guaranteed. The stability of the desired impedance dynamics plays an important role in human–robot cooperative motion through the convergence of the desired-trajectory tracking error when the interaction force is zero. The stability of impedance errors implies the achievement of the desired impedance dynamics by the designed impedance controllers. In constant-impedance control [4], the desired impedance profiles, including the desired stiffness, damping, and stiffness, are usually chosen as positive definite diagonal matrices that guarantee the stability of the desired impedance dynamics. Compared with constant-impedance control, variable impedance control provides more flexibility to change impedance profiles for PHRI; however, the desired variable impedance dynamics (DVID) in it is a time-varying linear system whose stability cannot be guaranteed by the positive definite and diagonal assumptions. In [5] and [6], the asymptotical stability of the DVID is guaranteed by novel constraints on time-varying impedance profiles. The impedance profiles constraints proposed in [7] ensure the exponential stability of the DVID. Robotic modeling uncertainty is the main fact that affects impedance errors performances. To date, various impedance control methods have been proposed to improve impedance errors performances based on adaptive control [8], [9], [10], [11], [12], [13], sliding mode techniques [14], [15], iterative learning [16], [17], and neural networks [18], [19], [20].

On human impedance profiles learning, various methods were proposed based on least-squares techniques, recursive least squares, dynamic movement primitives (DMPs) [21], [22], [23], and programming by demonstration [24], [25], [26]. The estimated impedance profiles have been directly used in impedance control [22], [23], [24], [25]; however, the time-varying impedance profiles may violate the stability of the DVID. In [27] and [28], adaptive optimal impedance profiles were obtained by solving a linear quadratic regulator (LQR) problem using adaptive dynamic programming in the outer loop, while adaptive (neural) controllers were designed in the inner loop to achieve the DVID.

It should be noted that almost all the abovementioned variable impedance controllers require real-time interaction forces. However, human–robot interaction force measurements

Manuscript received 21 March 2022; revised 21 August 2022 and 17 December 2022; accepted 4 February 2023. Date of publication 17 February 2023; date of current version 6 August 2024. This work was supported in part by the National Key Research and Development Project under Grant 2019YFB1312500; in part by the National Natural Science Foundation of China under Grant 62073156 and Grant 62103280; in part by the Beijing Science and Technology Program under Grant Z211100007921021; in part by the State Key Laboratory of Management and Control for Complex Systems, Institute of Automation, Chinese Academy of Sciences; and in part by the Shanghai Engineering Research Center of Assistive Devices. (*Corresponding author: Jiantao Yang.*)

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Digital Object Identifier 10.1109/TNNLS.2023.3243091

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may become impossible owe to no appropriate force sensors. Although interaction forces can be estimated through observers, the estimation is achieved by neglecting robot modeling uncertainties. Moreover, the measurement of interaction forces is usually latter than PHRI, which affects the reliability of the measured forces.

Without interaction force, model-based impedance learning controllers were developed [29], [30], [31], [32] for PHRI, where periodic adaptation-based impedance learning was proposed in [29] and [30], iterative impedance learning was developed in [31], and differential adaptive impedance learning was given in [32]. The impedance learning control approach is appealing for its control stability guarantee and its ability in providing variable impedance regulation for robots through online impedance learning without the requirement of interaction force sensing. Nevertheless, the impedance learning controllers in [29], [30], [31], and [32] have the following limitations: 1) the closed-loop control systems are only uniformly ultimately bounded (UUB), not asymptotically stable and 2) precise knowledge about robotic dynamic parameters is required in [29] and [30], and initial repositioning at each iteration is needed in [31]. In addition, the control implementation in [29], [30], [31], and [32] requires environmental impedance profiles being periodic, iteration-independent, or slowly varying. The slow variation requirements of impedance profiles restrict the applications of adaptive impedance learning control in [32], while impedance profiles in repetitive PHRI tasks cannot be completely periodic or iteration-independent, and there always exist impedance profiles differences in different task completion.

Repetitive PHRI tasks are common in robot-assisted rehabilitation therapies, leg exoskeletons, and some other cooperative repetition applications. For repetitive tasks, iterative learning control and repetitive learning control are appealing for their good control performances. Compared with iterative learning control, repetitive learning control can be applied to solve the finite-time tracking problem without initial repositioning at each cycle [33], [34]. To overcome the mentioned limitations in the last paragraph, this article proposes a novel model-based repetitive impedance learning control strategy for PHRI with repetitive cooperative tasks. The proposed control, which is composed of a proportional-differential (PD) control term, an adaptive control term, and an impedance learning term, can provide variable impedance regulation for repetitive PHRI tasks without interactive force sensing. Compared with the related results [29], [30], [31], [32], the contributions and innovations of this article are stated as follows.

- 1) This article proposes a differential adaptation law for estimating a constant vector of robotic parameters in the time domain and repetitive impedance learning laws for estimating time-varying impedance in the iterative domain.
- 2) Uniform convergence of tracking errors is guaranteed by PD control and the use of projection in the adaptive estimation and full saturation in the repetitive impedance learning laws and is theoretically proved based on a Lyapunov-like analysis and illustrated by simulations

on a parallel robot. The use of the projection and full saturation ensures the estimators being contained in related prescribed sets and avoids possible estimators drift.

- 3) In impedance profiles, stiffness and damping are composed of an iteration-independent term and an iteration-dependent disturbance, which are estimated by repetitive learning and compressed by the PD control, respectively. Therefore, the developed approach can be applied to the PHRI where iteration-dependent disturbances exist in the stiffness and damping.
- 4) Initial repositioning of the robotic system at each cycle is avoided by setting the initial value of the robot state and the adaptive estimator at the current iteration as the end value at the last iteration.

The rest of this article is organized as follows. Section II presents the robot dynamics. Section III gives the impedance learning-based repetitive learning control strategy and its control stability analysis. Simulation results are presented in Section IV. At last, conclusions are presented in Section V.

II. ROBOT DYNAMICS

Consider the robotic dynamics in Cartesian coordinates with the following form:

$$M(x)\ddot{x} + C(x, \dot{x})\dot{x} + G(x) = \tau + \tau_e \quad (1)$$

where $x, \dot{x}, \ddot{x} \in R^n$ denote the robotic position, velocity, and acceleration, respectively; $M(x) \in R^{n \times n}$ is the inertial matrix; $C(x, \dot{x}) \in R^{n \times n}$ is the Coriolis and centrifugal matrix; $G(x) \in R^n$ is the gravity torque; $\tau \in R^n$ denotes the robot control input, and $\tau_e \in R^n$ denotes the PHRI force.

The robotic Cartesian dynamics has the following properties [35].

Property 1: $M(x)$ is a symmetric and positive definite matrix, and

$$\sigma_1 I \leq M(x) \leq \sigma_2 I \quad (2)$$

where I denotes the identity matrix, and σ_1 and σ_2 are positive constants.

Property 2: $\dot{M}(x) - 2C(x, \dot{x})$ is skew symmetric, i.e.,

$$\xi^T (\dot{M}(x) - 2C(x, \dot{x})) \xi = 0 \quad \forall \xi \in R^n. \quad (3)$$

Property 3: The robotic dynamics in (1) can be parameterized as follows:

$$M(x)\phi_1 + C(x, \dot{x})\phi_2 + G(x) = W(\phi_1, \phi_2, x, \dot{x})\theta \quad (4)$$

where θ is a constant vector that contains unknown robotic parameters, and $\|\theta\| \leq \theta_c$ with θ_c being a positive constant. In rigid-link robots, the robotic parameters in θ include the length, the mass, and inertial moments of links.

Property 4: If $x, \dot{x} \in L_\infty$, then $M(x), C(x, \dot{x}), G(x)$, and their partial derivatives with respect to their respective arguments are bounded. If $\phi_1, \phi_2, x, \dot{x} \in L_\infty$, then $W(\phi_1, \phi_2, x, \dot{x})$ and its partial derivatives with respect to its arguments are bounded.

Remark 1: The dynamics in (1) with Properties 1–4 can describe the models of a range of robotics, such as robot arms, rehabilitation robots, exoskeletons, and surgical robots.

Define the trajectory tracking error e_k as follows:

$$e_k = x_r - x_k \quad (5)$$

where k denotes the k th iteration, and x_r denotes the desired trajectory with $x_r, \dot{x}_r, \ddot{x}_r \in L_\infty$.

The interactive force $\tau_{e,k}$ can be expanded as follows:

$$\tau_{e,k} = -K_k^*(t)e_k - B_k^*(t)\dot{e}_k \quad (6)$$

where $K_k^*(t)$ and $B_k^*(t)$ denote the iteration-dependent stiffness and damping, respectively. Furthermore, $K_k^*(t)$ and $B_k^*(t)$ can be expressed as follows:

$$\begin{aligned} K_k^*(t) &= K_h(t) + K_{d,k}(t) \\ B_k^*(t) &= B_h(t) + B_{d,k}(t) \end{aligned} \quad (7)$$

where $K_h(t)$ and $B_h(t)$ are iteration-independent terms that play a dominant role in $K_k^*(t)$ and $B_k^*(t)$, respectively; $K_{d,k}(t)$, $B_{d,k}(t)$ denote bounded iteration-dependent disturbances that satisfy $\|K_{d,k}(t)\| \leq c_k$ and $\|B_{d,k}(t)\| \leq c_b$ with $c_k, c_b > 0$.

Based on the definition of e_k , $\tau_{e,k}$ can be expressed as follows:

$$\tau_{e,k} = F_0(t) + K_h(t)x_k + B_h(t)\dot{x}_k - K_{d,k}(t)e_k - B_{d,k}(t)\dot{e}_k \quad (8)$$

where $F_0(t) := -K_h(t)x_r - B_h(t)\dot{x}_r$ denotes a feedforward term.

Remark 2: In practical applications, there always exist constant matrices f^* , K^* , and D^* , such that

$$\text{abs}(F_0(t)) \leq F^*, \quad \text{abs}(K_h(t)) \leq K^*, \quad \text{abs}(B_h(t)) \leq B^* \quad (9)$$

where $\text{abs}(\cdot)$ is the absolute value function, and each element on the left part of “ $\leq$ ” is simply the related element on the right part of “ $\leq$ ”.

Remark 3: In some PHRI [32], a feedforward human force $\tau_{f,k}$ is added in (6). As illustrated in [36], the feedforward force $\tau_{f,k}$ can be described by a function of x_r that is iteration-independent. Therefore, $\tau_{f,k}$ is iteration-independent, and (8) is a general form of the interactive force $\tau_{e,k}$.

Assumption 1: The desired trajectory $x_r(t) \in C^2$ and $x_r(0) = x_r(T)$, $\dot{x}_r(0) = \dot{x}_r(T)$.

Assumption 2: $x_k(0) = x_{k-1}(T)$ and $\dot{x}_k(0) = \dot{x}_{k-1}(T)$, where $k = 0, 1, 2, \dots$ denotes the iteration numbers.

III. REPETITIVE LEARNING CONTROL USING ITERATIVE IMPEDANCE LEARNING

To obtain the convergence of e_k , this section presents the proposed repetitive impedance learning controller, which is composed of a PD control term, an adaptive control term with projection modification, and a repetitive impedance learning term with full saturation.

Define the following filtered tracking error:

$$r_k = \dot{e}_k + k_1 e_k \quad (10)$$

where k_1 is a positive gain. Based on the dynamics in (1), r_k has the following dynamics:

$$\begin{aligned} M(x_k)\dot{r}_k &= W(\ddot{x}_r + k_1\dot{e}_k, \dot{x}_k + x_r, x_k, \dot{x}_k) \\ &\quad - C(x_k, \dot{x}_k)r_k - \tau_k - \tau_{e,k}. \end{aligned} \quad (11)$$

Design the control input τ_k (see Fig. 1 for the control schematics) as follows:

$$\tau_k = k_2 r_k + W(\ddot{x}_r + k_1\dot{e}_k, \dot{x}_k + x_r, x, \dot{x}_k)\hat{\theta}_k - \hat{\tau}_{e,k} \quad (12)$$

where $k_2 > 0$, and $\hat{\theta}_k$ and $\hat{\tau}_{e,k}$ are estimators of θ and $\tau_{e,k}$, respectively. Then, the dynamics of r_k can be represented as follows:

$$\begin{aligned} M(q_k)\dot{r}_k &= -C(q_k, \dot{q}_k)r_k + W(\ddot{q}_r + k_1\dot{e}_k, \dot{q}_k + q_r, q_k, \dot{q}_k)\tilde{\theta}_k \\ &\quad - k_2 r_k - \tilde{\tau}_{e,k} \end{aligned} \quad (13)$$

where $\tilde{\theta}_k = \theta - \hat{\theta}_k$ and $\tilde{\tau}_{e,k} = \tau_e - \hat{\tau}_{e,k}$ denote estimation errors.

Denote $\hat{F}_k(t)$, $\hat{K}_{h,k}$, and $\hat{B}_{h,k}$ as the estimators of $F_0(t)$, $K_h(t)$, and $B_h(t)$, respectively, at the k th cycle. Then, $\hat{\tau}_{e,k}$ can be designed as follows:

$$\hat{\tau}_{e,k} = \hat{F}_k(t) + \hat{K}_{h,k}(t)x_k + \hat{B}_{h,k}(t)\dot{x}_k \quad (14)$$

and the estimation error $\tilde{\tau}_{e,k}$ is

$$\begin{aligned} \tilde{\tau}_{e,k} &= \tilde{F}_k(t) + \tilde{K}_{h,k}(t)x_k + \tilde{B}_{h,k}(t)\dot{x}_k \\ &\quad - K_{d,k}(t)e_k - B_{d,k}(t)\dot{e}_k \end{aligned} \quad (15)$$

where $\tilde{F}_k(t) = F_0(t) - \hat{F}_k(t)$, $\tilde{K}_{h,k}(t) = K_{h,k}(t) - \hat{K}_{h,k}(t)$, and $\tilde{B}_{h,k}(t) = B_{h,k}(t) - \hat{B}_{h,k}(t)$.

The estimator $\hat{\theta}_k$ is updated by the following differential adaptive law:

$$\dot{\hat{\theta}}_k = \text{proj}(\gamma_0 W^T(\ddot{x}_r + k_1\dot{e}_k, \dot{x}_r + k_1e_k, x_k, \dot{x}_k)r_k) \quad (16)$$

$$\hat{\theta}_k(0) = \hat{\theta}_{k-1}(T), \quad \hat{\theta}_0(0) = 0 \quad (17)$$

where $\gamma_0 > 0$ is the learning rate, and $\text{proj}(\bullet)$ is a projection function defined as follows:

$$\text{proj}(\bullet) = \begin{cases} \bullet, & \text{if } \|\hat{\theta}_k\| \leq c_\theta \\ \bullet - \frac{\hat{\theta}_k \hat{\theta}_k^T \bullet}{\|\hat{\theta}_k\|^2}, & \text{if } \|\hat{\theta}_k\| = c_\theta \text{ \& } \hat{\theta}_k^T \bullet \leq 0 \\ \bullet - \frac{\hat{\theta}_k \hat{\theta}_k^T \bullet}{\|\hat{\theta}_k\|^2}, & \text{if } \|\hat{\theta}_k\| = c_\theta \text{ \& } \hat{\theta}_k^T \bullet > 0. \end{cases} \quad (18)$$

The estimators $\hat{F}_k(t)$, $\hat{K}_{h,k}$, and $\hat{B}_{h,k}$ are iteratively updated using full saturation as follows:

$$\begin{aligned} (i) \quad \hat{F}_k(t) &= \text{sat}_{F^*}(\hat{F}_k(t)) \\ \hat{F}_k(t) &= \hat{F}_{k-1}(t) - \gamma_1 r_k(t) \end{aligned} \quad (19)$$

$$\begin{aligned} (ii) \quad \hat{K}_{h,k}(t) &= \text{sat}_{K^*}(\hat{K}_{h,k}(t)) \\ \hat{K}_{h,k}(t) &= \hat{K}_{h,k-1}(t) - \gamma_2 r_k(t)x_k^T(t) \end{aligned} \quad (20)$$

$$\begin{aligned} (iii) \quad \hat{B}_{d,k}(t) &= \text{sat}_{B^*}(\hat{B}_{d,k}(t)) \\ \hat{B}_{h,k}(t) &= \hat{B}_{h,k-1}(t) - \gamma_3 r_k(t)\dot{x}_k^T(t) \end{aligned} \quad (21)$$

for $t \in [0, T]$, where $\hat{F}_k(t)$, $\hat{K}_{h,k}(t)$, and $\hat{B}_{h,k}(t)$ are respective auxiliary matrices, and r_1, r_2 , and r_3 are repetitive learning rates designed as positive definite diagonal matrices. In applications, the learning rates are designed based on trial and evaluation using PHRI performance.

In (19), $\text{sat}_{F^*}(\cdot)$ denotes the saturation function defined by

$$\text{sat}_{F^*}(\hat{F}_{ki}) = \begin{cases} \hat{F}_{ki}, & \text{if } |\hat{F}_{ki}| \leq F_i^* \\ \hat{F}_{ki} \text{sgn}(\hat{F}_{ki}), & \text{otherwise.} \end{cases} \quad (22)$$

The functions $\text{sat}_{K^*}(\cdot)$ and $\text{sat}_{B^*}(\cdot)$ are defined in a manner similar to the definition of $\text{sat}_{F^*}(\cdot)$.

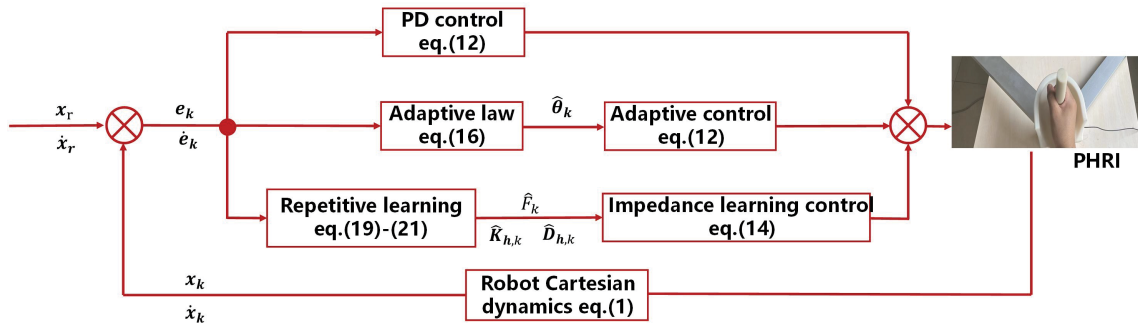


Fig. 1. Schematics of the proposed repetitive impedance learning-based physically human-robot interactive controller.

Remark 4: The use of the projection modification in (16) and the full saturations in (19)–(21) can guarantee the estimators being contained in prescribed sets. This avoids parameter drift in differential adaptation-based estimation and avoids the divergence in repetitive learning due to parameters accumulation in the same direction. The bound of $\text{abs}(F_0(t))$, $\text{abs}(K_h(t))$, and $\text{abs}(B_h(t))$ is usually not exactly known. In practical applications, the bound can be chosen relatively large in the estimators' initial implementation based on prior knowledge. In further implementation, the bound can be reduced to some extent based on estimation performance.

Remark 5: It is a good choice by using the differential update law to estimate the time-invariant vector θ in the time domain and using repetitive learning for estimating the time-varying impedance profiles in the iterative domain. The reasons are stated as follows: 1) the time-varying impedance profiles cannot be effectively estimated by using differential update laws for their inherent deficiencies in estimating time-varying signals; 2) the differential update law is more suitable for estimating the time-invariant vector θ , since differential updating mechanism may generate a smoother profile comparing with the difference type; and 3) estimating the time-invariant vector θ and the time-varying impedance profiles in different domains can decrease the uncertainties coupling to some extent.

Theorem 1: For the robot with the dynamics in (1), design the repetitive impedance learning-based physically human-robot interactive control strategy in (12), where $k_2 - c_b - 1/k_1 - c_k^2/k_1 - k_1 c_b^2$, $\hat{\theta}_k$ is updated by the differential type updating law in (16), and $\tilde{F}_k(t)$, $\tilde{K}_{h,k}(t)$, and $\tilde{B}_{d,k}(t)$ are updated by the repetitive learning laws in (19)–(21). Then, the tracking error e_k uniformly converges to zero over $[0, T]$, and $\tilde{\theta}_k$, $\tilde{F}_k(t)$, $\tilde{K}_{h,k}(t)$, and $\tilde{B}_{h,k}(t)$ are uniformly bounded over $[0, T]$ by properly choosing the control gains.

Proof: Consider the following composite energy functions:

$$E_k(t) = V_k(t) + P_k(t) \quad (23)$$

where $V_k(t)$ is defined as follows:

$$V_k(t) = \frac{1}{2} e_k^T e_k + \frac{1}{2} r_k^T M(x_k) r_k + \frac{1}{2} \tilde{\theta}_k^T \gamma_0^{-1} \tilde{\theta}_k \quad (24)$$

$$P_k(t) = \frac{1}{2} \int_0^t \tilde{F}_k^T \gamma_1^{-1} \tilde{F}_k d\tau + \frac{1}{2} \int_0^t \text{tr}\{\tilde{K}_{h,k}^T \gamma_2^{-1} \tilde{K}_{h,k}\} d\tau + \frac{1}{2} \int_0^t \text{tr}\{\tilde{B}_{h,k}^T \gamma_3^{-1} \tilde{B}_{h,k}\} d\tau. \quad (25)$$

A. Part A—Difference of Composite Energy Functions

$$\begin{aligned} \Delta V_k(t) &= V_k(t) - V_{k-1}(t) \\ &= \int_0^t \frac{dV_k}{d\tau} d\tau + V_k(0) - V_{k-1}(t) \\ &= \frac{1}{2} \int_0^t r_k^T (\dot{M}(x_k) - 2C(x_k, \dot{x}_k)) r_k d\tau + \int_0^t e_k^T \dot{e}_k d\tau \\ &\quad - \int_0^t r_k^T \tilde{\tau}_{e,k} d\tau - \int_0^t \tilde{\theta}_k^T \gamma_0^{-1} (\dot{\hat{\theta}}_k - r_0 W_k^T r_k) d\tau \\ &\quad - k_2 \int_0^t r_k^T r_k d\tau + V_k(0) - V_{k-1}(t) \end{aligned} \quad (26)$$

where W_k denotes $W(\ddot{q}_r + k_1 \dot{e}_k, \dot{q}_k + q_r, q_k, \dot{q}_k)$ for simplicity.

Since $\dot{M}(x_k) - 2C(x_k, \dot{x}_k)$ is skew symmetric, $r_k^T (\dot{M}(x_k) - 2C(x_k, \dot{x}_k)) r_k = 0$. The projection modification in (16) guarantees that the following hold: 1) $\|\hat{\theta}_k\| \leq c_\theta$ for $t \geq 0$ if $\|\hat{\theta}_k(0)\| \leq c_\theta$ and 2) $\tilde{\theta}_k^T \gamma_0^{-1} (\dot{\hat{\theta}}_k - \gamma_0 W_k^T r_k) \geq 0$. Since (17) holds, then $\|\hat{\theta}_k\| \leq c_\theta$ always holds for $t \in [0, T]$. Applying these results into (26) yields

$$\begin{aligned} \Delta V_k(t) &\leq \int_0^t [r_k^T (-k_2 r_k + K_{d,k}(t) e_k + B_{d,k}(t) \dot{e}_k) + e_k^T \dot{e}_k] d\tau \\ &\quad - \int_0^t [r_k^T \tilde{F}_k + \text{tr}\{\tilde{K}_{h,k}^T r_k x_k^T\} + \text{tr}\{\tilde{K}_{h,k}^T r_k \dot{x}_k^T\}] d\tau \\ &\quad + V_k(0) - V_{k-1}(t). \end{aligned} \quad (27)$$

Based on the definition of r_k , one can get $e_k^T \dot{e}_k = -k_1 e_k^T e_k + e_k^T r_k$ and $r_k^T [K_{d,k}(t) e_k + B_{d,k}(t) \dot{e}_k] = r_k^T B_{d,k}(t) r_k + r_k^T (K_{d,k}(t) - k_1 B_{d,k}(t)) e_k$. Using the Young's inequality

$$\begin{aligned} r_k^T e_k &\leq k_1 e_k^T e_k / 4 + r_k^T r_k / k_1 \\ r_k^T K_{d,k} e_k &\leq k_1 e_k^T e_k / 4 + c_k^2 r_k^T r_k / k_1 \\ r_k^T k_1 B_{d,k} e_k &\leq k_1 e_k^T e_k / 4 + k_1 c_b^2 r_k^T r_k. \end{aligned} \quad (28)$$

Substituting these results into (27), one can obtain

$$\begin{aligned} \Delta V_k(t) &\leq \int_0^t [-\bar{k}_2 r_k^T r_k - k_1 / 4 e_k^T e_k] d\tau - \int_0^t r_k^T \tilde{F}_k d\tau \\ &\quad - \int_0^t [\text{tr}\{\tilde{K}_{h,k}^T r_k x_k^T\} - \text{tr}\{\tilde{B}_{h,k}^T r_k \dot{x}_k^T\}] d\tau \\ &\quad + V_k(0) - V_{k-1}(t) \end{aligned} \quad (29)$$

where $\bar{k}_2 = k_2 - c_b - 1/k_1 - c_k^2/k_1 - k_1 c_b^2 > 0$.

The difference of $\tilde{F}_k^T \tilde{F}_k$ is as follows:

$$\begin{aligned} \tilde{F}_k^T \tilde{F}_k - \tilde{F}_{k-1}^T \tilde{F}_{k-1} &= (\hat{F}_k - \hat{F}_{k-1})^T (\hat{F}_k + \hat{F}_{k-1} - 2F_0) \\ &= -2\tilde{F}_k^T (\hat{F}_k - \hat{F}_{k-1}) - \|\hat{F}_k - \hat{F}_{k-1}\|^2 \\ &= -2\tilde{F}_k^T (\hat{F}_k - \hat{F}_{k-1}) + 2\tilde{F}_k^T (\hat{F}_k - \hat{F}_k) - \|\hat{F}_k - \hat{F}_{k-1}\|^2. \end{aligned} \quad (30)$$

From [37, Lemma 2.1]

$$(F_0 - \text{sat}_{F^*}(\hat{F}_k))^T (\hat{F}_k - \text{sat}_F(\hat{F}_k)) \leq 0. \quad (31)$$

Applying (31) into (30), one can obtain

$$\begin{aligned} \tilde{F}_k^T \tilde{F}_k - \tilde{F}_{k-1}^T \tilde{F}_{k-1} &\leq -2\tilde{F}_k^T (\hat{F}_k - \hat{F}_{k-1}) \\ &= 2\tilde{F}_k^T \gamma_1 r_k. \end{aligned} \quad (32)$$

Similarly, one can get

$$\begin{aligned} \text{tr}\{\tilde{K}_{h,k}^T \tilde{K}_{h,k}\} - \text{tr}\{\tilde{K}_{h,k-1}^T \tilde{K}_{h,k-1}\} &\leq 2\text{tr}\{\tilde{K}_{h,k}^T \gamma_2 r_k x_k^T\} \\ \text{tr}\{\tilde{B}_{h,k}^T \tilde{B}_{h,k}\} - \text{tr}\{\tilde{B}_{h,k-1}^T \tilde{B}_{h,k-1}\} &\leq 2\text{tr}\{\tilde{B}_{h,k}^T \gamma_3 r_k \dot{x}_k^T\}. \end{aligned} \quad (33)$$

Combining the above analysis, one can get

$$\Delta E_k(t) \leq \int_0^t [-\bar{k}_2 r_k^T r_k - k_1/4 e_k^T e_k] d\tau + V_k(0) - V_{k-1}(t). \quad (34)$$

From Assumptions 1 and 2, $r_k(0) = r_{k-1}(T)$, $\tilde{\theta}_k(0) = \tilde{\theta}_{k-1}(T)$, and $V_k(0) = V_{k-1}(T)$. Then, from (34)

$$\Delta E_k(T) \leq -k_2 \int_0^T r_k^T(\tau) r_k(\tau) d\tau - \frac{k_1}{4} \int_0^T e_k^T(\tau) e_k(\tau) d\tau. \quad (35)$$

B. Part B—Convergence in L_T^2 Sense

The finiteness of $E_k(T)$ is required to show the convergence of r_k in L_T^2 sense. From (34), the finiteness of $E_k(T)$ can be ensured by the finiteness of $E_0(T)$, which is validated in the subsequent part.

Based on the analysis in Part A, the time derivative of $E_0(t)$ satisfies

$$\begin{aligned} \dot{E}_0(t) &\leq -k_2 r_0^T r_0 - r_0^T \tilde{F}_0 - \text{tr}\{\tilde{K}_0^T r_0 \dot{q}_0^T\} - \text{tr}\{\tilde{B}_0^T r_0 \dot{q}_0^T\} \\ &\quad + 1/2 \tilde{F}_0^T \gamma_1^{-1} \tilde{F}_0 + 1/2 \text{tr}\{\tilde{K}_0^T \gamma_2^{-1} \tilde{K}_0\} \\ &\quad + 1/2 \text{tr}\{\tilde{B}_0^T \gamma_3^{-1} \tilde{B}_0\}. \end{aligned} \quad (36)$$

Based on the iterative update laws in (19)–(21)

$$\frac{1}{2} \tilde{F}_0^T \gamma_1^{-1} \tilde{F}_0 \leq r_0^T \tilde{F}_0 + \frac{1}{2} F_0^T \gamma_1^{-1} F_0 \quad (37)$$

$$\frac{1}{2} \text{tr}\{\tilde{K}_0^T \gamma_2^{-1} \tilde{K}_0\} \leq \text{tr}\{\tilde{K}_0^T r_0 x_0^T\} + \frac{1}{2} \text{tr}\{K_h^T \gamma_1^{-1} K_h\} \quad (38)$$

$$\frac{1}{2} \text{tr}\{\tilde{B}_0^T \gamma_3^{-1} \tilde{B}_0\} \leq \text{tr}\{\tilde{B}_0^T r_0 \dot{x}_0^T\} + \frac{1}{2} \text{tr}\{B_h^T \gamma_3^{-1} B_h\}. \quad (39)$$

Since $F_0(t)$, $K_h(t)$, and $B_h(t)$ are bounded, from (36)–(39), one can see that $\dot{E}_0(t)$ is bounded; i.e., there exists a positive constant c_0 , such that $|\dot{E}_0(t)| \leq c_0$. Furthermore

$$E_0(t) \leq E_0(0) + \int_0^t |\dot{E}_0(\tau)| d\tau \leq E_0(0) + c_0 T < \infty \quad (40)$$



Fig. 2. Render graph of the considered planar parallel robot.

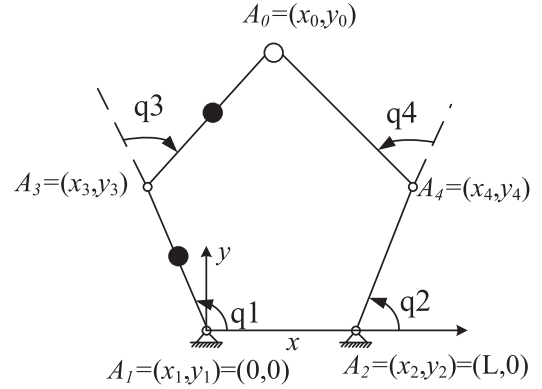


Fig. 3. Structure diagram of the planar parallel robot.

for $t \in [0, T]$. From (35) and (40), $E_k(T)$ is bounded for $k = 0, 1, 2, \dots$

According to (35), one can get

$$\begin{aligned} E_k(T) &\leq E_0(T) - k_2 \sum_{j=1}^k \int_0^T \|r_j(\tau)\|^2 d\tau \\ &\quad - \frac{k_1}{4} \sum_{j=1}^k \int_0^T e_k^T(\tau) e_k(\tau) d\tau \end{aligned} \quad (41)$$

$$\begin{aligned} \lim_{k \rightarrow \infty} E_k(T) &\leq E_0(T) - k_2 \lim_{k \rightarrow \infty} \sum_{j=1}^k \int_0^T \|r_j(\tau)\|^2 d\tau \\ &\quad - \frac{k_1}{4} \lim_{k \rightarrow \infty} \sum_{j=1}^k \int_0^T e_k^T(\tau) e_k(\tau) d\tau. \end{aligned} \quad (42)$$

Since $E_k(T) \geq 0$ and $E_0(T)$ is bounded, according to the convergence theorem of the sum of series, one can obtain the convergence of r_k and e_k in L_T^2 sense, i.e.,

$$\lim_{k \rightarrow \infty} \int_0^T \|r_k(\tau)\|^2 d\tau = \lim_{k \rightarrow \infty} \int_0^T \|e_k(\tau)\|^2 d\tau = 0. \quad (43)$$

C. Part C—Uniform Convergence

From the definition of $E_k(t)$ and (34)

$$\begin{aligned} E_k(t) &= \Delta E_k(t) + E_{k-1}(t) \\ &\leq \int_0^t [-\bar{k}_2 r_k^T r_k - k_1/4 e_k^T e_k] d\tau + V_k(0) \\ &\quad + P_{k-1}(t). \end{aligned} \quad (44)$$

The finiteness of $E_k(T)$ implies $V_k(T) \in L_\infty$. As $V_k(0) = V_{k-1}(T)$, $V_k(0) \in L_\infty$. From the iterative learning using full

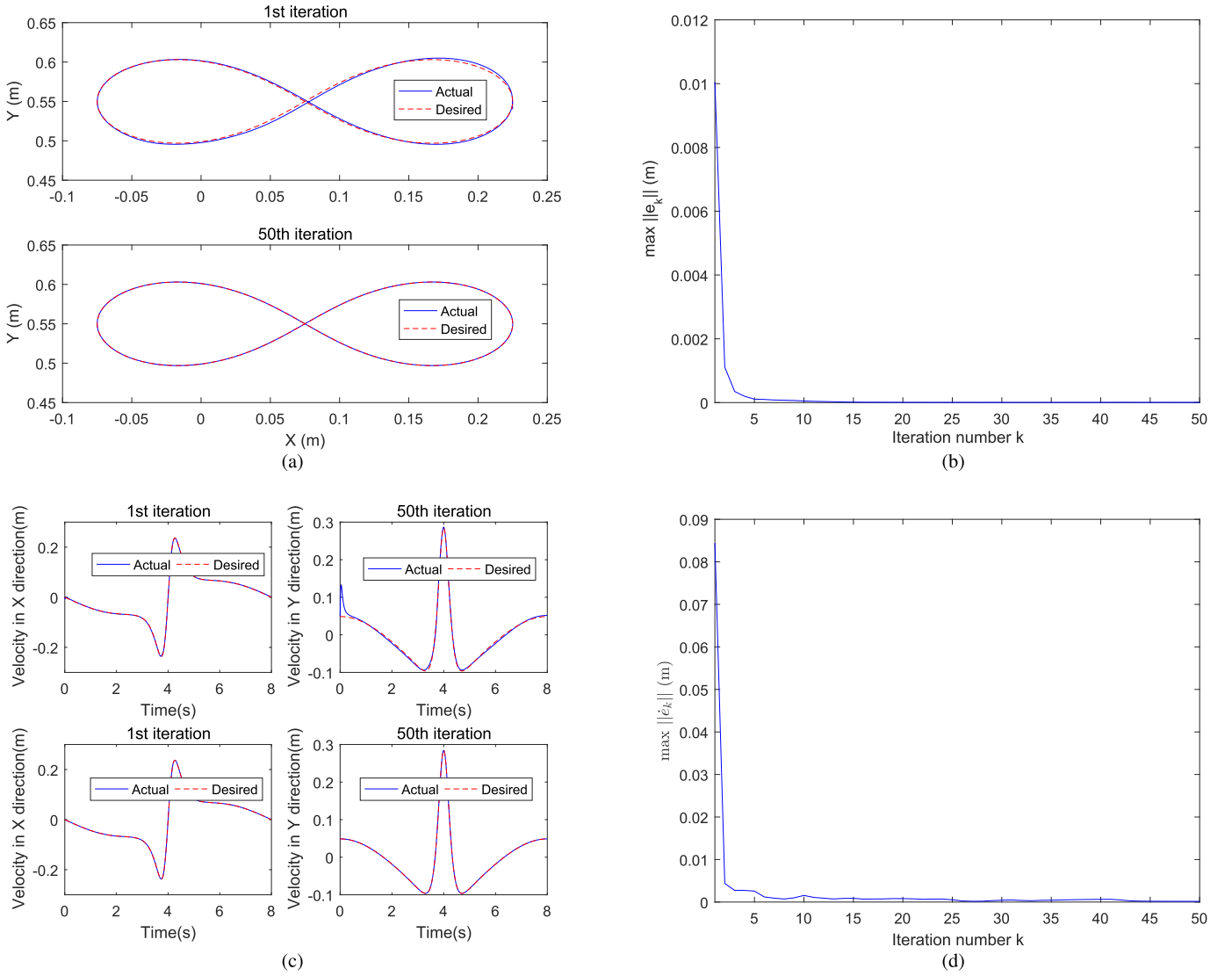


Fig. 4. Tracking performance and maximum tracking error by the proposed iterative impedance learning-based repetitive learning controller. (a) Position tracking performance at the first and 30th iterations. (b) Maximum position tracking error at each iteration. (c) Velocity tracking performance at the first and 50th iterations. (d) Maximum velocity tracking error at each iteration.

saturations in (19)–(21), $\tilde{F}_k(t)$, $\tilde{K}_{h,k}(t)$, $\tilde{B}_{h,k}(t) \in L_\infty$, which implies $P(t) \in L_\infty$ for $t \in [0, T]$. From $V_k(0) \in L_\infty$ and $P_{k-1}(t) \in L_\infty$, one can conclude that $E_k(t) \in L_\infty$, which implies $e_k(t), r_k(t) \in L_\infty$. Based on the finiteness of $e_k(t), r_k(t)$ and the definition of $r_k(t), \dot{e}_k(t) \in L_\infty$, which implies that $e_k(t)$ is uniformly continuous. From the uniform continuity and the convergence in L_2^T sense of $x_k(t)$, the uniform convergence of e_k can be concluded.

Remark 6: As repetitive learning is used in the control design, the proposed iterative impedance learning-based repetitive learning control can be applied in robot-assisted rehabilitation, leg exoskeletons, and other cooperative repetitive tasks.

Remark 7: In the proposed impedance learning-based repetitive learning control, the initial repositioning in iterative learning control is avoided because of the following: 1) the closed characteristics of the desired trajectory and 2) the initialization of the robot dynamics and the robot parameters estimator at the k th iteration as the final related values at the

TABLE I
PARAMETERS OF THE PARALLEL ROBOT

Link	Mass(kg)	Length(m)	Inertial moment(kg.m ²)
A_2A_4	$m_2 = 0.3$	$l_2 = 0.5$	$I_2 = 0.1$
A_4A_0	$m_4 = 0.2$	$l_4 = 0.5$	$I_4 = 0.075$
A_0A_3	$m_3 = 0.2$	$l_3 = 0.5$	$I_3 = 0.075$
A_3A_1	$m_1 = 0.3$	$l_1 = 0.5$	$I_1 = 0.1$

$k - 1$ th iteration. To be specific, Assumptions 1 and 2 and the initialization in (17) ensure $V_k(0) = V_{k-1}(T)$ based on which the import result in (35) is obtained for control stability analysis.

IV. SIMULATIONS

Parallel robot has been widely used as service robots for its features, including high stiffness, compact structure, large bearing capacity, and good dynamic response. In this article, we illustrate the effectiveness and advantages of the proposed

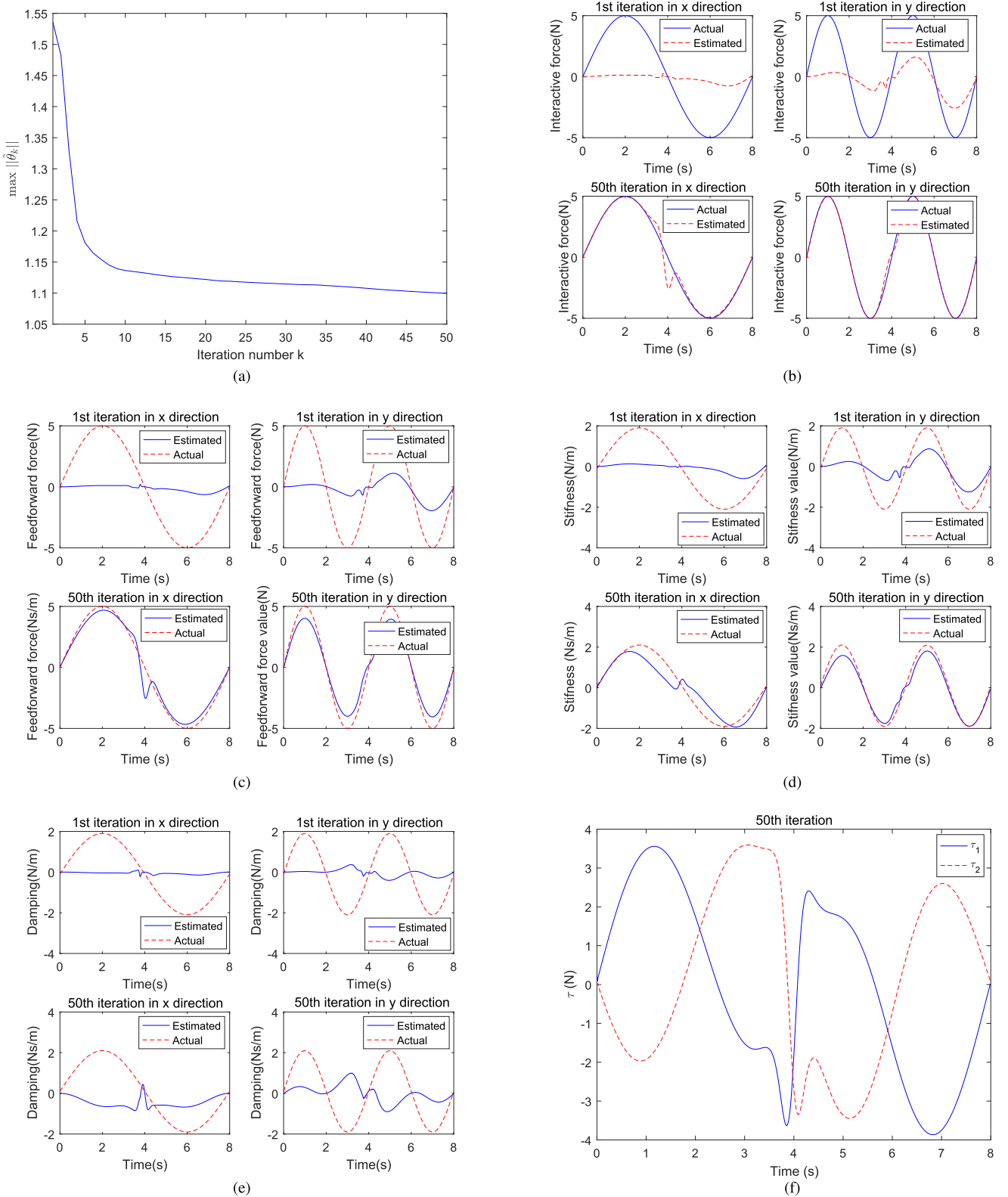


Fig. 5. Estimation performance and the control input by the proposed iterative impedance learning-based repetitive learning controller. (a) Maximum norm $\|\hat{\theta}_k\|$ at each iteration. (b) Estimation performance of $\hat{f}_k$ at the first and 50th iterations. (c) Feedforward force estimation at the first and 50th iterations. (d) Stiffness estimation performance at the first and 50th iterations. (e) Damping estimation performance at the first and 50th iterations. (f) Control input at the 50th iteration.

PHRI control on a planar parallel robot with the render graph (see Fig. 2) in a five-bar parallel structure (Fig. 3). For the parallel robot, the values of A_i , $i = 0, 1, 2, 3, 4$, are five joint

points, where A_0 is the HRI point, and A_2A_4 , A_4A_0 , A_0A_3 , and A_3A_1 denote four robotic links whose masses, lengths, and inertial moments are give in Table I. The distances between

the joints and respective centers of mass are all 0.25 m, i.e., $l_{ci} = 0.25$ m for $i = 1, 2, 3, 4$, and $L = 0.15$ m denotes the length of A_1A_2 . The joint space dynamics of the parallel robot is modeled based on the Newton–Euler method in which an acceleration analysis is realized by the influence coefficient method. Then, the joint space dynamics is converted into the Cartesian dynamics of the parallel robot based on their relationship. By calculation, parameterization of the robot Cartesian dynamics is obtained with the constant vector θ being given in the Appendix.

To repetitively follow the Lemniscate of Bernoulli: $(x^2 + y^2)^2 = a(x^2 - y^2)$ with $a = 0.075\sqrt{2}$, the considered parallel robot requires to repetitively track the reference trajectory $x_r = [x_{1r}, x_{2r}]^T$ given by:

$$x_{1r} = 0.075 + \frac{3\sqrt{2}\cos(\omega t) + \cos(2\omega t) + 3}{2\sqrt{2}\cos(\omega t) + 3}a \quad (45)$$

$$x_{2r} = 0.55 + \frac{\sqrt{2}\sin(\omega t) + \sin(2\omega t)}{2\sqrt{2}\cos(\omega t) + 3}a \quad (46)$$

where $\omega = 2\pi/T$ with $T = 8$ as the time period. During the PHRI, let the iteration-independent stiffness and damping as $K_h = B_h = 2\text{diag}\{\sin(\pi t/4), \sin(\pi t/2)\}$, the iteration-dependant stiffness and damping as $K_{d,k} = B_{d,k} = 0.1\cos(\pi t/k)I$, and the feedforward force as $F_0(t) = 5\text{diag}\{\sin(\pi t/4), \sin(\pi t/2)\}$.

In the simulation, the parameters vector θ represents the robotic constant uncertainty, while the human feedforward force, the stiffness, and the damping represent time-varying impedance uncertainties. As accurate robotic dynamic parameters and iteration-independent stiffness and damping are required in [29] and [30], the model-based impedance learning controllers in [29] and [30] cannot be implemented. To show the control effectiveness, for the proposed iterative impedance learning-based repetitive learning control, design the following control parameters: $k_1 = 10$, $k_2 = 80$, $\gamma_0 = 120 I$, $\gamma_1 = 200 I$, $\gamma_2 = \text{diag}\{450, 120\}$, and $\gamma_3 = 400 I$. To illustrate the control advantages, comparisons are taken with the adaptive PHRI control in [32] with the control parameters: $\zeta = 10$, $\sigma = 80$, $\Lambda_k = 700 I$, $\Psi_k = 20 I$, $\Lambda_B = 600 I$, $\Psi_B = 10 I$, $\Lambda_f = 600 \text{diag}\{600, 300\}$, $\Psi_f = \text{diag}\{20, 10\}$, $\Lambda_\theta = 120 I$, $\Psi_\theta = 12$, $\Lambda_\eta = 80 I$, and $\Psi_\eta = 8 I$.

Simulation results by the proposed repetitive impedance learning-based physically human–robot interactive controller are presented in Figs. 4 and 5 with the iteration period $T = 8$ s and a total simulation time of 400 s. From Fig. 4, the proposed control makes the robot trajectory and velocity coincide with the desired ones and renders the tracking errors e_k and $\dot{e}_k$ converge to zero after the 45th iteration. Based on this, the proposed impedance learning control can provide variable impedance regulation for PHRI with uniformly asymptotical stability guarantee. From Fig. 5(a), the maximum estimation error $\|\tilde{\theta}_k\|$ decreases a lot after ten iterations, which shows the effectiveness of the proposed adaptation law in estimating the constant vector of robotic parameters. From Fig. 5, the estimation of the interactive force, the feedforward force, and the stiffness obtains satisfactory performance at the 30th iteration in comparison with that at the first iteration, which

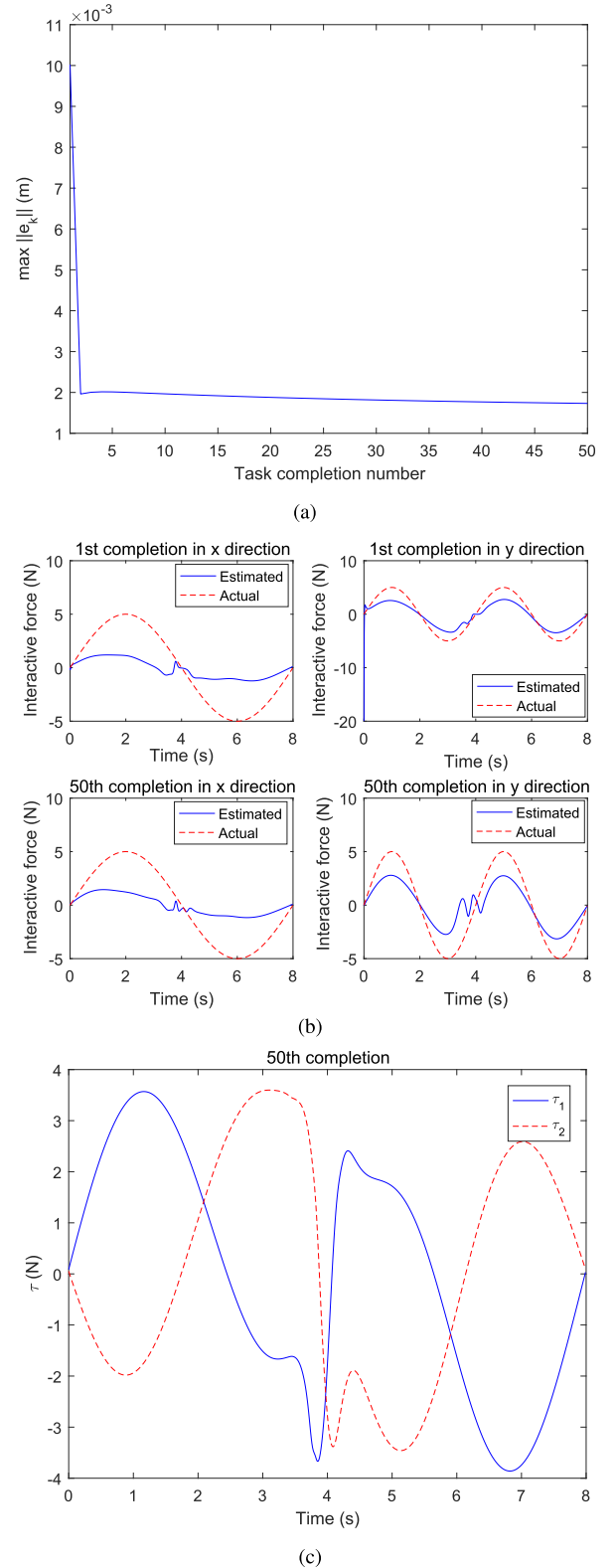


Fig. 6. Performance by the adaptive PHRI control in [32]. (a) Maximum position tracking error at each task completion. (b) Performance of the interactive force estimation at the first and 50th task completion. (c) Control input at the 50th task completion.

illustrates the effectiveness of the proposed fully saturated repetitive learning laws in (19)–(21). It should be noted that the estimators cannot coincide with the related actual values in

Fig. 5(b)–(e) owing to the coupling among the robotic parameters uncertainties and the impedance uncertainties. In addition, there exist mutations in Fig. 5(b)–(f) at about the fourth second, because there is a mutation in the desired velocity at this time when the Lemniscate of Bernoulli has a cross point.

The proposed model-based impedance learning control can guarantee the uniform convergence of the tracking error, while the adaptive PHRI in [32] only ensures the boundedness of the tracking error shown in Fig. 6(a). Comparing Figs. 5(b) and 6(b), the designed fully saturated repetitive learning laws obtain satisfactory performance in estimating the time-varying interactive force, while the estimation performance of the interactive force in Fig. 6(b) is not satisfactory owe to the inherent deficiencies of the adaptive learning laws [32] in estimating time-varying signals. Therefore, the proposed impedance learning control has a better performance of the tracking error than the adaptive PHRI in [32], and the designed fully saturated repetitive learning laws have better performance in estimating the considered time-varying interactive force than differential adaptation laws.

V. CONCLUSION

In this study, a novel repetitive impedance learning control approach is proposed for PHRI in repetitive tasks. The proposed control overcomes the limitations of the related model-based impedance learning controllers and guarantees the uniform convergence of tracking errors for PHRI with robotic parameters uncertainties and time-varying impedance uncertainties. The stability of the closed-loop control system is proven based on a Lyapunov-like analysis, and the control effectiveness and advantages are illustrated by simulations on a parallel robot. It should be noting that the developed approach can be applied to the PHRI where iteration-dependent disturbances exist in the stiffness and damping. Future work will focus on model-based impedance learning control based on spatial repetitive learning using the spatial periodicity characteristics of the repetitively followed path.

APPENDIX

The vector θ in (5) for the considered robot dynamics is with the following elements:

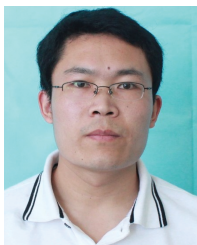
$$\begin{aligned}\theta_1 &= \frac{l_1 l_3^2 l_4^2}{l_3^2 l_4^2}; & \theta_2 &= \frac{l_3 l_1^2 l_4^2}{l_3^2 l_4^2}; & \theta_3 &= \frac{l_4 l_1^2 l_3^2}{l_3^2 l_4^2} \\ \theta_4 &= \frac{l_1^2 l_3^2 l_4^2 m_3}{l_3^2 l_4^2}; & \theta_5 &= \frac{l_3^2 l_4^2 l_1^2 m_1}{l_3^2 l_4^2}; & \theta_6 &= \frac{l_1^2 l_4^2 l_3^2 m_3}{l_3^2 l_4^2} \\ \theta_7 &= \frac{l_1^2 l_3^2 l_4^2 m_4}{l_3^2 l_4^2}; & \theta_8 &= \frac{l_1^2 l_3 l_4^2 l_3 m_3}{l_3^2 l_4^2}; & \theta_9 &= \frac{l_2 l_3^2 l_4^2}{l_3^2 l_4^2} \\ \theta_{10} &= \frac{l_3 l_2^2 l_4^2}{l_3^2 l_4^2}; & \theta_{11} &= \frac{l_4 l_2^2 l_3^2}{l_3^2 l_4^2}; & \theta_{12} &= \frac{l_2^2 l_3^2 l_4^2 m_4}{l_3^2 l_4^2} \\ \theta_{13} &= \frac{l_3^2 l_4^2 l_2^2 m_2}{l_3^2 l_4^2}; & \theta_{14} &= \frac{l_2^2 l_4^2 l_3^2 m_3}{l_3^2 l_4^2}; & \theta_{15} &= \frac{l_2^2 l_3^2 l_4^2 m_4}{l_3^2 l_4^2} \\ \theta_{16} &= \frac{l_2^2 l_3 l_4 l_4 m_4}{l_3^2 l_4^2}; & \theta_{17} &= \frac{l_1 l_2 l_4 l_3^2}{l_3^2 l_4^2}; & \theta_{18} &= \frac{l_1 l_2 l_3 l_4^2}{l_3^2 l_4^2}\end{aligned}$$

$$\begin{aligned}\theta_{19} &= \frac{l_1 l_2 l_3^2 l_4^2 m_4}{l_3^2 l_4^2}; & \theta_{20} &= \frac{l_1 l_2 l_4^2 l_3^2 m_3}{l_3^2 l_4^2}; & \theta_{21} &= \frac{l_1 l_2 l_3 l_4^2 l_3 m_3}{l_3^2 l_4^2} \\ \theta_{22} &= \frac{l_1 l_2 l_3^2 l_4 l_4 m_4}{l_3^2 l_4^2}.\end{aligned}$$

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